

Screening biochemical changes induced by psychedelic ayahuasca in healthy subjects and patients with depression: a machine learning approach

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Abstract

In a secondary analysis of a double-blind, parallel-arm, randomized placebo-controlled trial, we investigated Ayahuasca, an ancient psychedelic beverage from Amazonia with potential antidepressant effects. This study assessed its impact on biochemical markers in healthy subjects (control group - CG, n = 42) and patients with treatment-resistant depression (TRD, n = 28). Using machine learning techniques, specifically random forest algorithms with recursive feature elimination and 5-fold cross-validation, our model identified c-reactive protein (CRP), awakening salivary cortisol, and total cholesterol as key features differentiating CG and TRD. The model showed low performance in distinguishing Ayahuasca from placebo in the CG, suggesting minimal impact on these biomarkers in healthy subjects. In contrast, for TRD, the model indicated modulation by Ayahuasca in glucose, CRP, lymphocytes, LDL cholesterol, red blood cells, BDNF, and AST. Recursive feature elimination and correlation analysis highlighted BDNF and plasmatic cortisol as predictors of Ayahuasca's therapeutic outcomes in TRD. Our findings reinforce Ayahuasca's safety in healthy individuals and associate biomarker shifts with its effect in TRD patients.

Introduction

Depression is a public health issue that affects approximately 5% of the global population[1]. Notably, nearly 30% of those suffering from major depressive disorders may experience treatment-resistant depression (TRD), where standard treatments are ineffective[2][3]. Besides, many of those who respond to antidepressants still have to wait for weeks or months to feel the effects of the medication, while many experience significant side effects[4]. In the last years, psychedelics such as psilocybin, MDMA, ayahuasca, and LSD gained interest in psychiatry, which spurred a new wave of research and clinical trials for major depression and other mental diseases[5][6].

The results are promising. Current evidence suggests that psychedelics have good tolerability, safety, and low toxicity[7][8]. Phase III clinical trials with MDMA assisted-therapy suggested a therapeutic effect in patients with post-traumatic stress disorder[9]. Phase IIb clinical trials with psilocybin suggest rapid and sustained antidepressant effects[10][11]. Our research group conducted two clinical trials with ayahuasca, an Amerindian psychedelic brew, for patients with TRD who did not respond to at least two antidepressants from different classes. An open-label trial observed that a single dose of ayahuasca produced significant antidepressant effects in TRD[12]. A randomized placebo-controlled trial found significant antidepressant effects of ayahuasca when compared with placebo at one and two days ($p = 0.04$), and seven days ($p < 0.0001$) after dosing, with the response rate on the order of 64%[13].

We have also explored the potential mechanisms behind the therapeutic effects of ayahuasca on depression. Our evidence suggests that the phenomenology of the psychedelic acute effects plays an important role in the antidepressant effects[14]. Additionally, changes in specific biochemical markers were associated with the observed antidepressant effects, which include serum Brain-Derived Neurotrophic Factor (BDNF), serum cortisol, and plasma C-Reactive Protein (CRP), pointing toward increased markers of neuroplasticity[15][16], as well as reductions on markers of inflammation[17] and stress[17].

In the present study, we sought to answer four questions concerning the effects of Ayahuasca on a panel of biochemical markers. a) Do these biomarkers distinguish between healthy individuals and patients with Treatment-Resistant Depression (TRD)? b) Which of these biomarkers would best predict the dosing session with ayahuasca on healthy subjects? c) Which of these biomarkers would best predict an ayahuasca dosing session on TRD patients? d) Do changes in specific biomarkers correlate with ayahuasca's observed antidepressant effect in TRD? Such analysis will allow us to better understand the relationships between multiple variables simultaneously, which may help us to identify patterns or connections between physiological systems in response to ayahuasca that would not be detected using univariate analysis.

Methods

This study was a secondary analysis of a double-blind, parallel-arm, randomized placebo-controlled trial for both healthy individuals and patients with treatment-resistant depression. The primary outcomes of this trial, which strictly adhered to the CONSORT 2010 guidelines[18], are detailed in [12]. Participants were recruited through media and Internet advertisements, or psychiatrist referrals from local outpatient psychiatric units. To improve blinding, all participants were naive to ayahuasca. The clinical trial procedures took place at the Onofre Lopes University Hospital (HUOL), Federal University of Rio Grande do Norte (UFRN), Natal, Brazil. The study was approved by the HUOL Research Ethics Committee (# 579.479). All subjects provided written informed consent before participation. This study was first registered on 29/11/2013 at ClinicalTrials.gov (NCT02914769) and was conducted in accordance with the Declaration of Helsinki.

Subjects

Healthy controls: We selected 42 healthy adults naive to ayahuasca, ensuring they met the following inclusion criteria: no current medical disease based on history, physical examination, and routine hematologic and biochemical tests; no current or previous history of psychiatric or neurological disorders; no abusive use of substances; and no pregnancy.

Patients: We selected 28 adults who met the criteria and were in a current episode of unipolar major depressive disorder as diagnosed by the Structured Clinical Interview for Axis I (DSM-IV). All selected patients were treatment-resistant depressed, who had unsatisfactory responses to at least two different classes of antidepressant medications. Exclusion criteria were: previous experience with ayahuasca; current medical disease based on history, physical examination, and routine hematologic and biochemical tests; current or previous history of neurological disorders, psychosis, schizophrenia, or bipolar affective disorder; history of mania or hypomania; use of substances of abuse; suicidal risk; and pregnancy.

Substances

Ayahuasca: We used a single batch of ayahuasca prepared and provided free of charge by a division of the Barquinha church based in the city of Ji-Paraná, Brazil. The ayahuasca beverage used in the RCT consisted of (mean $\pm$ SD): 0.36 ± 0.01 mg/mL of DMT, 1.86 ± 0.11 mg/mL of harmine, 0.24 ± 0.03 mg/mL of harmaline, and 1.20 ± 0.05 mg/mL of tetrahydroharmine [12]. Dosing was prepared per participant as 1 mL (ayahuasca or placebo) / kg (body weight) [12].

Placebo: To improve the blinding of the substance, we used a placebo preparation designed to mimic a few properties of ayahuasca, with a brownish color, bitter, and sour taste. It was prepared with water, citric acid, yeast, zinc sulfate, and a caramel colorant. Zinc sulfate is bitter and induces mild gastrointestinal distress, such as nausea, vomiting, and sometimes diarrhea, which are the most well-known effects of ayahuasca.

Procedures

Participants slept in the lab the night before dosing for sleep baseline assessment. They were awakened around 6 am after about 7 hours of sleep. At 7 am, volunteers were instructed to collect saliva by themselves. Blood samples were drawn from the participants by a trained nurse. A baseline clinical evaluation was conducted by a trained psychiatrist, including the evaluation of depression severity with the Montgomery-Asberg Depression Rating Scale (MADRS). The dosing sessions lasted approximately 8 hours, between 8 am and 4 pm. Participants were given either ayahuasca (n total = 36; 22 controls and 14 patients) or placebo (n total = 34; 20 controls and 14 patients). The experiments took place in the university hospital, in a quiet and comfortable living room-like environment, with a bed, a recliner, controlled temperature, and dimmed light. The substance intake occurred around 10 am. All volunteers received a single dose of ayahuasca or placebo. Participants were comfortably accommodated in a recliner. They were offered the use of headphones to listen to a playlist created for the study [13]. They were asked to remain quiet, with their eyes closed (no eye shades) while focusing their attention on their body, thoughts, and emotions. They were instructed to try keeping with specific attitudes and intentions [14]. Two researchers from the team were sitting next (~3 m) to the participants, guiding the procedure, operating an electroencephalogram and electrocardiogram system, and offering support to the participant when needed. At the end of the session, participants were debriefed on their experience, had a final psychiatric evaluation, and at around 5 pm they were allowed to go home accompanied by a relative or a friend. One day after the dosing session, all participants returned to the hospital for a new round of testing. After a night of sleep in the lab (D2), biological samples were collected at rest, around 7 am, and MADRS was assessed.

Biological samples collection and dosage

Saliva samples were collected immediately upon waking up, 30 and 45 minutes after, using a specific cotton stick called Salivette (Sarstedt, Germany). Before and during collection, subjects remained at rest and no liquid or food was allowed. The area under the curve (AUC) was calculated from the 3 points. The larger cubital vein was punctured using a needle and a 20 ml syringe to collect the blood subsequently divided into 4 ml tubes with EDTA and two 4 ml clot activator tubes. For biological samples collection, all volunteers were fasting for at least 8 hours, and at complete rest. Blood and saliva analyses were performed at the Laboratory of Clinical Analysis (HUOL) and Laboratory of Hormonal Measures (UFRN). For biochemical analysis, the blood and saliva were centrifuged (3500 rpm for 10 minutes) and the supernatant was retained for dosages. We included 31 biochemical markers (features) in the analysis: 1. alanine transaminase (ALT); 2. aspartate transaminase (AST); 3. creatinine (CRE); 4. urea (URE); 5. sodium ion (Na⁺); 6. potassium ion (K⁺); 7. High-Density Lipoproteins (HDL); 8. Low-Density Lipoproteins (LDL); 9. total cholesterol (CHOL); 10. triglycerides (TRIG); 11. glucose (GLU); 12. red blood cell count

(RBC); 13. hematocrit (HT); 14. hemoglobin (HB); 15. platelets (PLAT); 16. leukocytes (LEU); 17. lymphocytes in relative amounts (LYMPH%); 18. absolute lymphocytes (LYMPH); 19. monocytes in relative amounts (MONO%); 20. absolute monocytes (MONO); 21. neutrophils in relative amounts (NEU%); 22. absolute neutrophils (NEU); 23. neutrophils segmented in relative amounts (SEG%); 24. neutrophils segmented absolute (SEG); 25. eosinophils relative (EOS%); 26. eosinophils absolute (EOS); 27. C-reactive protein (CRP); 28. interleukin 6 (IL-6); 29. Brain-Derived Neurotrophic Factor (BDNF); 30. plasmatic cortisol (PLASMCORT); 31. salivary cortisol (SALCORT). Complete blood count (hematocrit, hemoglobin, leukocytes, lymphocytes, monocytes, neutrophils, platelets, RBC, segmented) was performed by flow cytometry with polarized light scattering using the XN-1000s equipment (Sysmex Corporation, Kobe, Japan). Concentrations of ALT, AST, creatinine, glucose, HDL, total cholesterol, triglycerides, and urea were measured by spectrophotometry. Sodium and potassium levels were quantified by ion-selective electrodes. Low-Density Lipoproteins (LDL) concentrations were calculated using Martin's formula. All measurements were performed using the CMD 800i X1 chemical analyzer (Wiener lab, Rosario, Argentina). CRP levels were dosed by immunoturbidimetry. BDNF, plasmatic and salivary cortisol, and IL-6 were assessed by ELISA as reported previously [12, 15, 16].

Machine Learning Approach

Pre-Processing

We first associated the biological measurements and clinical scores at baseline with labels that indicate whether data came from a TRD patient or a healthy individual. Labels were also used to identify the treatment (ayahuasca or placebo). We then calculated the differences between post-treatment (D2) and baseline values (referred to as 'delta'), i.e., we subtracted D2 values by baseline values.

Random Forests algorithms

The machine learning approach was conducted using Python scripts and the scikit-learn framework, which provided implementations of the Random Forest and Linear Regression algorithms, as well as the Cross-Validation and Recursive Feature Elimination (RFE) procedures. We assessed the potential of the collected blood and salivary variables (referred to as 'features' throughout the paper) to distinguish between a) TRD patients from healthy controls at baseline; b) healthy controls dosed with ayahuasca from those dosed with placebo (delta data analysis); c) TRD patients dosed with ayahuasca from those dosed with placebo (delta data analysis).

To develop the classification process, we initially trained a few machine learning models and compared the accuracy to predict whether the data came from a TRD patient or a healthy control (Table S1). Although Random Forest was not the top-performing model in terms of mean accuracy, it achieved competitive results and was chosen for the following reasons.

The Random Forest algorithm combines the output of multiple decision trees by adding random subsets, also known as bagging, which reduces the risk of overfitting and produces more accurate predictions[19]. It's a non-parametric model, and provides a direct measure of feature importance, allowing us to rank our variables based on their contribution to the model. Furthermore, Random Forest was capable of handling our dataset, which had a small number of observations relative to the number of features. This feature, combined with its robustness and flexibility, and its ability to provide important insights about the relevance of different features, made it a suitable choice for our analysis despite not having the highest mean accuracy in the initial comparison.

Finally, we also assessed the potential of selected features to predict the antidepressant effects measured by the MADRS scale (d). This part of the analysis was performed using Random Forest regression. We evaluated the potential of features to predict the antidepressant effects measured by the MADRS scale, specifically for patients dosed with ayahuasca.

To identify the most relevant features, we employed Recursive Feature Elimination with Cross-Validation (RFECV), a method that works by recursively eliminating features and building a model on those features that remain. It uses model accuracy to identify which features contribute the most to predict the target variable[20]. RFECV has the advantage of pruning a large feature space and enabling the construction of simpler, more interpretable models without compromising performance.

We performed RFECV with 5-fold cross-validation for classification tasks and 2-fold cross-validation for the regression task. The choice of 5-fold for classification was due to its balance between bias and variance, and it's a standard choice in many machine learning tasks. The choice of 2-fold for regression was due to the limited sample size, which could lead to an overestimation of the test error rate if a higher fold was used.

After performing RFECV, we further investigated the selected variables by conducting Pearson correlations. Specifically in d), we also used selected variables that showed significant correlations to create new models. To evaluate the effectiveness of our feature selection and model performance, we compared the performance of our models before and after feature selection. For the classification tasks, we calculated accuracy, specificity, sensitivity, and the area under the ROC curve (AUROC). For the regression task, we computed R^2 , mean absolute error (MAE), and root mean squared error (RMSE). These metrics provided insights into the impact of feature selection on the predictive power of the models and allowed for a more robust understanding of the data and the relationships therein.

Statistical analysis

Following the determination of the most relevant features by RFECV, we performed Wilcoxon tests on selected features selected as significant by the machine learning algorithms. Cross-checking the statistically significant variables allows for assessing the

robustness of the feature selection approach, as well as aids to understand if the effects are related to ayahuasca, placebo, or to both.

Large Language Model

We employed Python and R scripts for data analysis, developed and interpreted by the research team. To enhance these scripts, we used ChatGPT-4, an AI language model from OpenAI. While the AI-assisted in refining the scripts, all developmental and interpretive decisions were made by the authors.

Results

Biochemical differences between patients with TRD and healthy controls at baseline

Figure 1 shows the key features for distinguishing between TRD patients and healthy individuals at baseline. The heatmap in panel A illustrates the impact of each biomarker, with darker purple indicating higher significance in differentiation and light yellow lower significance.. Salivary cortisol, C-reactive protein, and total cholesterol stood out as the main features with substantial impact. Panel B displays the feature importance ranking, highlighting the selected features (awakening salivary cortisol, C-reactive protein, and total cholesterol) determined by the mean Gini decrease procedure. This ranking reinforces the significance of these features in the classification task.

Regarding the output of the classification task,, our models achieved an AUC ROC of 0.768, specificity of 0.474, sensitivity of 0.812, and accuracy of 0.671 using all features. Performance improved when using the selected features, resulting in an AUCROC of 0.850, specificity of 0.746, sensitivity of 0.812, and accuracy of 0.786 (Supplementary Figure 1).

Supplementary Figure 2 illustrates the results of the Wilcoxon tests conducted on the selected variables. The median awakening salivary cortisol level in the control group was 514.5 nmol/L, while in the TRD group, it was 327.0 nmol/L, indicating marginally significant decreased levels in TRD patients ($p = 0.09$). The median CRP level in controls was 1.00 mg/dL, whereas, in TRD patients it was significantly higher, 3.87 mg/dL ($p = 0.0013$). Similarly, the median total cholesterol level in the control group was 170.5 mg/dL, while in the TRD group, it was elevated to 209.0 mg/dL ($p = 0.0036$).

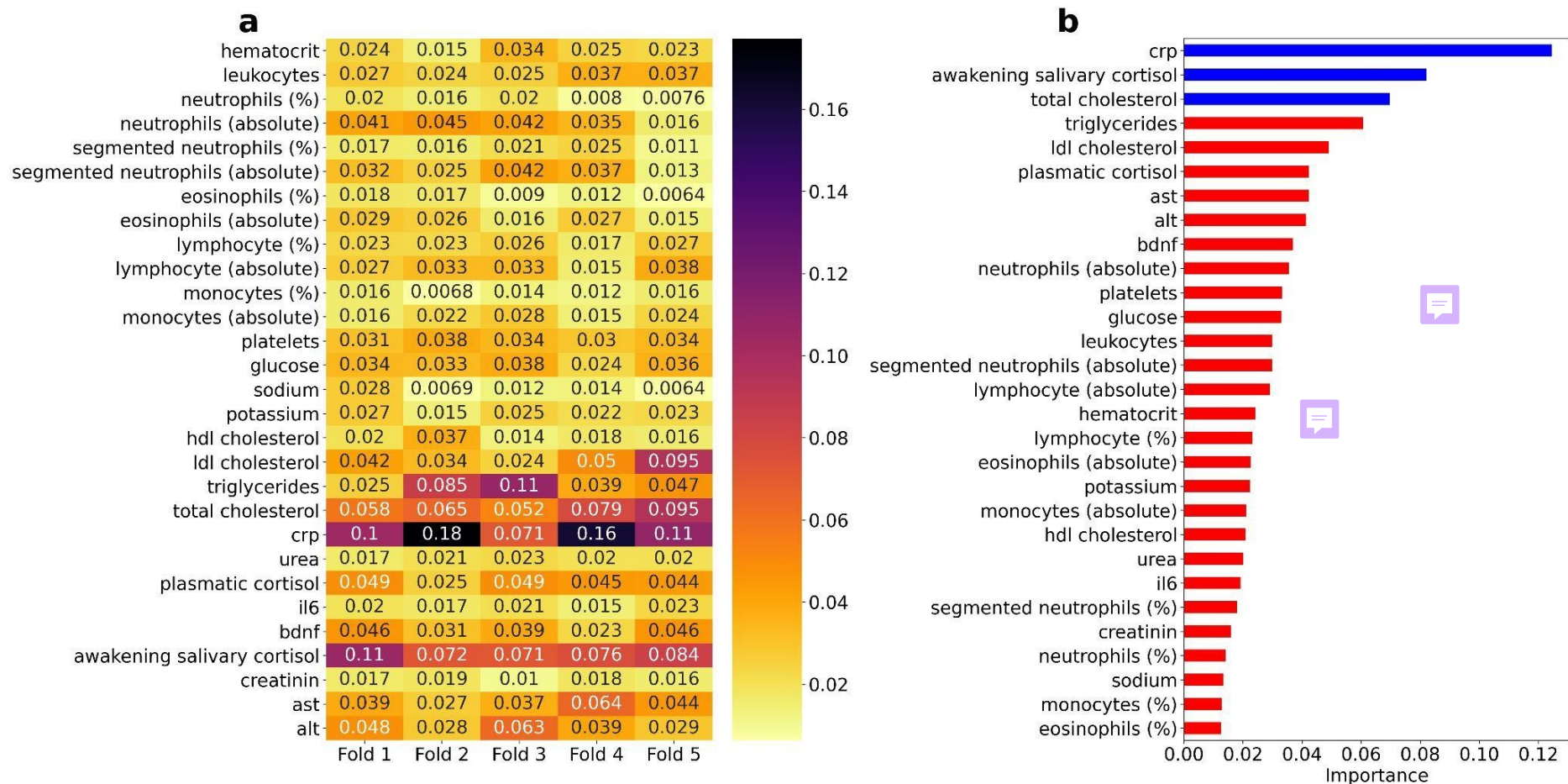


Figure 1. A) Heatmap of Gini indexes from a 5-Fold Cross-Validation procedure comparing TRD patients with healthy individuals (control group) at baseline. Values represent Mean Decrease Gini indicating each biomarker's importance. Purple and yellow represent high and low significance, respectively. Main features include salivary cortisol, C-reactive protein, and total cholesterol. B) Feature importance ranking of models. Features are ranked by mean Gini decrease, highlighting their importance in distinguishing TRD patients at baseline. The blue bars represent features selected by RFE: awakening salivary cortisol, C-reactive protein, and total cholesterol.

Biochemical differences between healthy controls dosed with ayahuasca or placebo

To differentiate between healthy controls dosed with ayahuasca or placebo, our machine learning model did not perform well. Considering all features, our model achieved an AUC ROC of 0.67, specificity of 0.5, sensitivity of 0.62, and accuracy of 0.56, while the model using the selected features showed an AUC ROC of 0.54, specificity of 0.5, sensitivity of 0.77, and accuracy of 0.64 (Supplementary Figure 1). These results suggest that ayahuasca may not significantly affect any of the biomarkers in the control group.

Biochemical differences in TRD patients dosed with ayahuasca or placebo

Our machine learning model performed well when differentiating between TRD patients dosed with ayahuasca or placebo. The original model achieved an AUCROC of 0.66, specificity of 0.432, sensitivity of 0.668, and accuracy of 0.54, while the model using the selected features demonstrated improved performance with an AUCROC of 0.9, specificity of 0.934, sensitivity of 0.734, and accuracy of 0.82 (Supplementary Figure 1). These results suggest that ayahuasca may have a significant impact on biomarkers in TRD patients.

In the TRD group, the machine learning model selected eight features using the RFE method, including glucose, CRP, lymphocyte (relative count), lymphocytes (absolute count), AST, BDNF, red blood cell count, and LDL cholesterol.

To further investigate the model's performance and its ability to distinguish between treatments, we performed Wilcoxon tests of the selected features (Supplementary Table 2). After dosing with placebo, there was a marginally significant decrease in median glucose levels (Median, before = 97, after = 93, $p = 0.0512$), while lymphocyte levels showed a significant increase (Median, before = 32, after = 35.5, $p = 0.0214$), suggesting an immune response. Meanwhile, after dosing with ayahuasca, a marginally significant decrease was observed in median Red Blood Cell (RBC) levels (Median, before = 4.60, after = 4.54, $p = 0.0685$). Furthermore, there was a significant reduction in median C-Reactive Protein (CRP) levels (Median, before = 5, after = 2.8, $p = 0.0056$), suggesting a potential reduction in inflammation.

Considering Aspartate Aminotransferase (AST), both the ayahuasca and placebo groups did not show significant changes. For the ayahuasca group, a slight increase was noticed (Median before: 19.50, after: 22.00, $p = 0.4139$). The placebo group, on the other hand, showed a slight decrease (Median before: 19.50, after: 18.00, $p = 0.8227$). Regarding Brain-Derived Neurotrophic Factor (BDNF), there were non-significant changes in both the ayahuasca and placebo groups. The ayahuasca group showed a decrease in median (Median before: 11815.50, after: 10495.00, $p = 0.2166$), while the placebo group showed a very slight decrease (Median before: 11454.00, after: 11380.50, $p = 0.8077$). In the case of Low-Density Lipoprotein Cholesterol (LDL), there were also non-significant increases in both the ayahuasca and placebo groups. For the ayahuasca group, a slight increase was observed (Median before: 145.00, after: 146.00, $p = 0.6375$). In the placebo group, a small increase was observed (Median before: 106.00, after: 110.00, $p = 0.2080$).

Even though AST, BDNF, and LDL cholesterol did not exhibit significant differences individually, their combined effects and potential nonlinear interactions with other features might have contributed to the model's ability to distinguish between treatments (ayahuasca or placebo). The random forest analysis is well-suited to capture nonlinear relationships and complex interactions, allowing for a more comprehensive understanding of the multifaceted nature of the relationship between treatments and TRD patients.

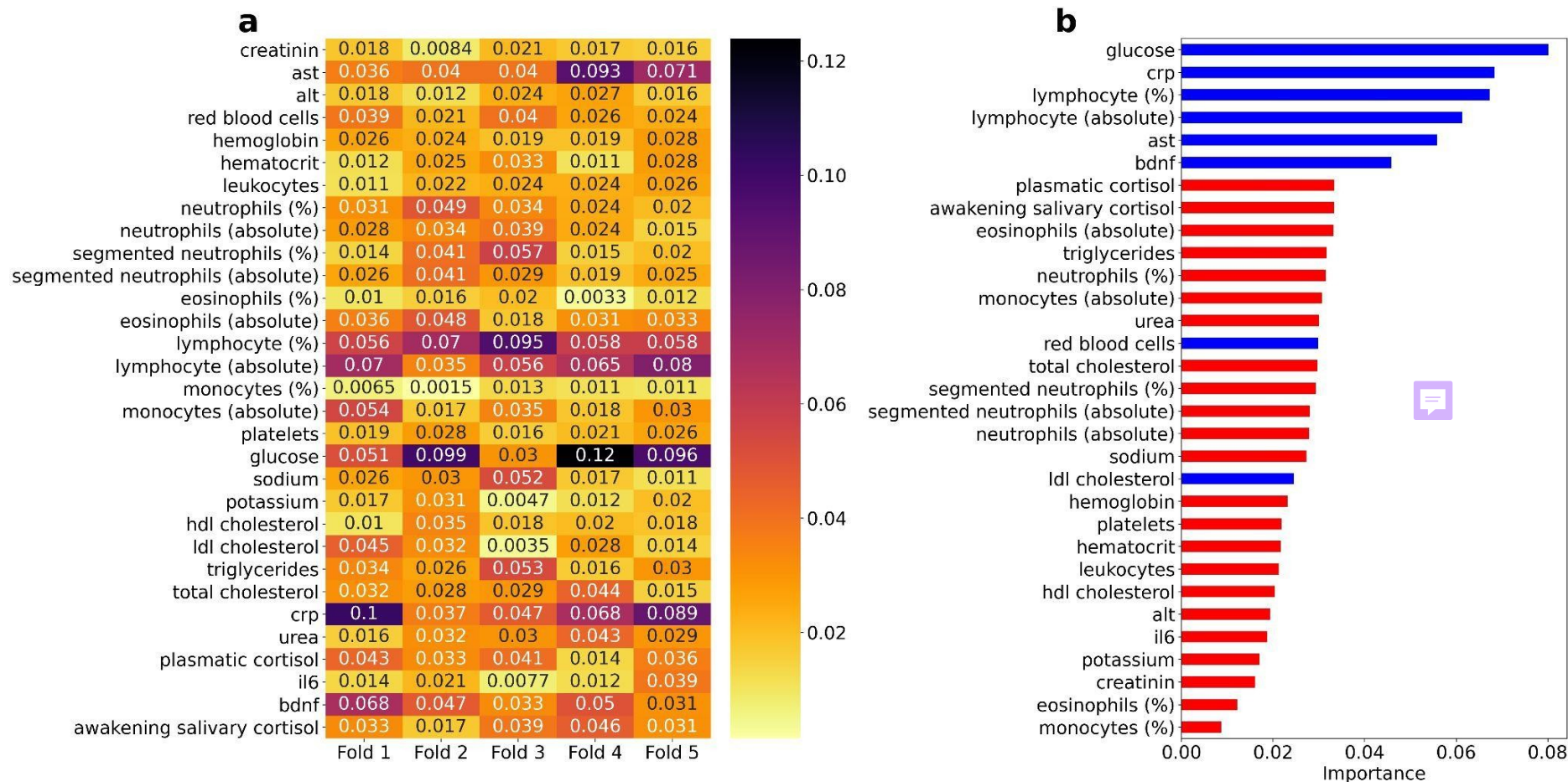


Figure 2. A) Heatmap of Gini indexes from a 5-Fold Cross-Validation procedure to classify depression patients dosed with ayahuasca or placebo. Values represent each biomarker's importance in distinguishing between groups. Black and purple denote more important features, while yellow and orange indicate less important ones. B) Feature importance ranking of models. Features are ranked by mean Gini decrease, showing their role in distinguishing TRD patients who used ayahuasca or placebo. The blue bars represent the top eight features selected by RFE: glucose, CRP, lymphocyte (relative and absolute count), AST, BDNF, red blood cell count, and LDL cholesterol.

The impact of the different features in predicting the clinical outcome in patients with TRD

In our study, we aimed to understand the impact of various biomarkers on the clinical outcomes in TRD patients. We utilized Recursive Feature Elimination with Cross-Validation (RFECV) in the Random Forest regression model to predict delta MADRS scores. The RFECV selected 19 biomarkers (Figure 3A) as significant predictors, providing insight into the potential of these variables in explaining the MADRS changes in response to ayahuasca treatment.

To find a statistically significant relationship between these variables and the predicted outcomes, we performed a correlation investigation using these selected variables (Supplementary Table 5). We found that BDNF and plasmatic cortisol had the most significant correlations with delta MADRS scores. Figure 3B depicts a strong negative correlation with BDNF (-0.6077 , $p = 0.02113$), and plasmatic cortisol (0.6014 , $p = 0.0228$, Fig. 3C).

Based on these findings, we constructed a new model containing only BDNF and plasmatic cortisol. Interestingly, this model showed improved predictive performance. The R^2 score increased from 0.82 for the RFECV model to 0.91 for the reduced model, while the mean R^2 score using all features was -0.09 . This highlights the improved performance of the reduced model compared to the original RFECV model and the model using all features.

We also evaluated the model's performance using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) metrics. For the model using all features, we found an RMSE of 5.8623 and an MAE of 4.2407. The model using the 19 features selected by RFECV yielded an RMSE of 17.3893 and an MAE of 12.8914. Remarkably, the reduced model containing only BDNF and plasmatic cortisol improved these metrics further, resulting in an RMSE of 12.6070 and an MAE of 9.7371.

These results underscore the superior predictive accuracy of the reduced model. The lower RMSE suggests that the reduced model provides predictions that are closer to the observed values, and the lower MAE indicates that the average absolute prediction error is minimized. Despite the simplicity of the reduced model, it outperformed the more complex models, demonstrating the power of feature selection in improving model performance and interpretability.

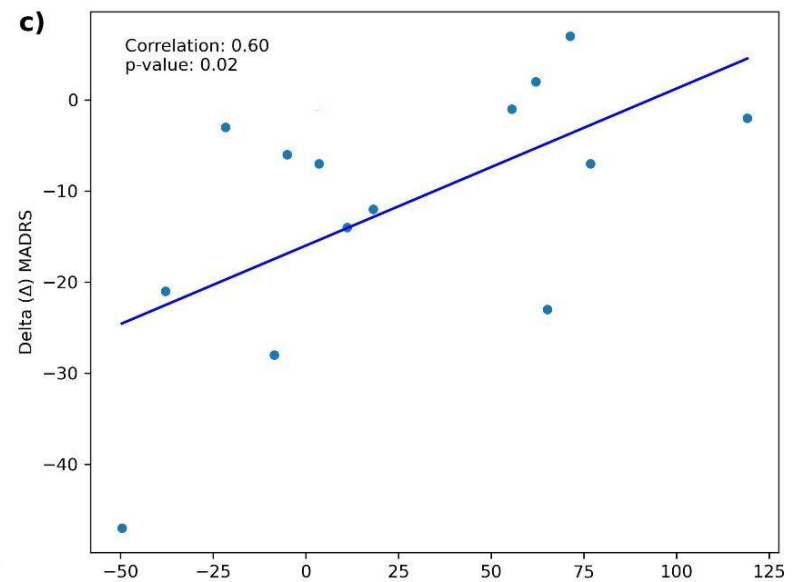
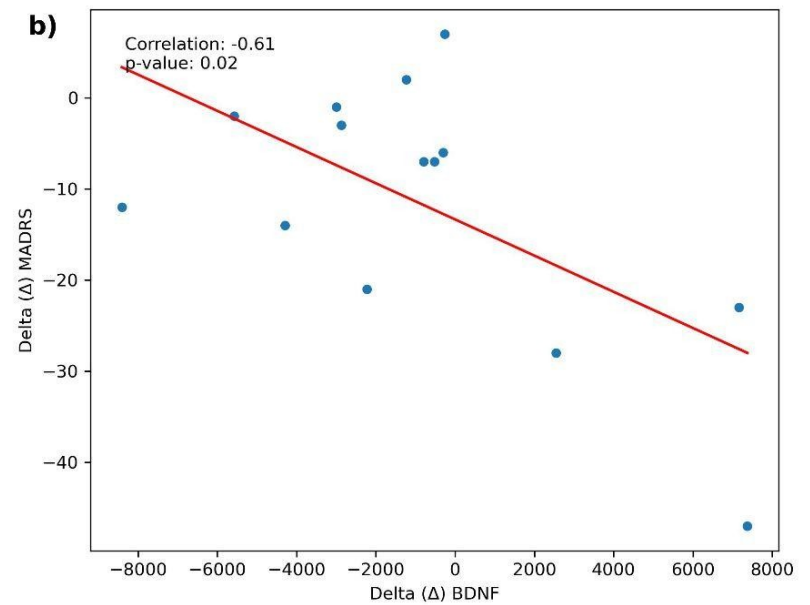
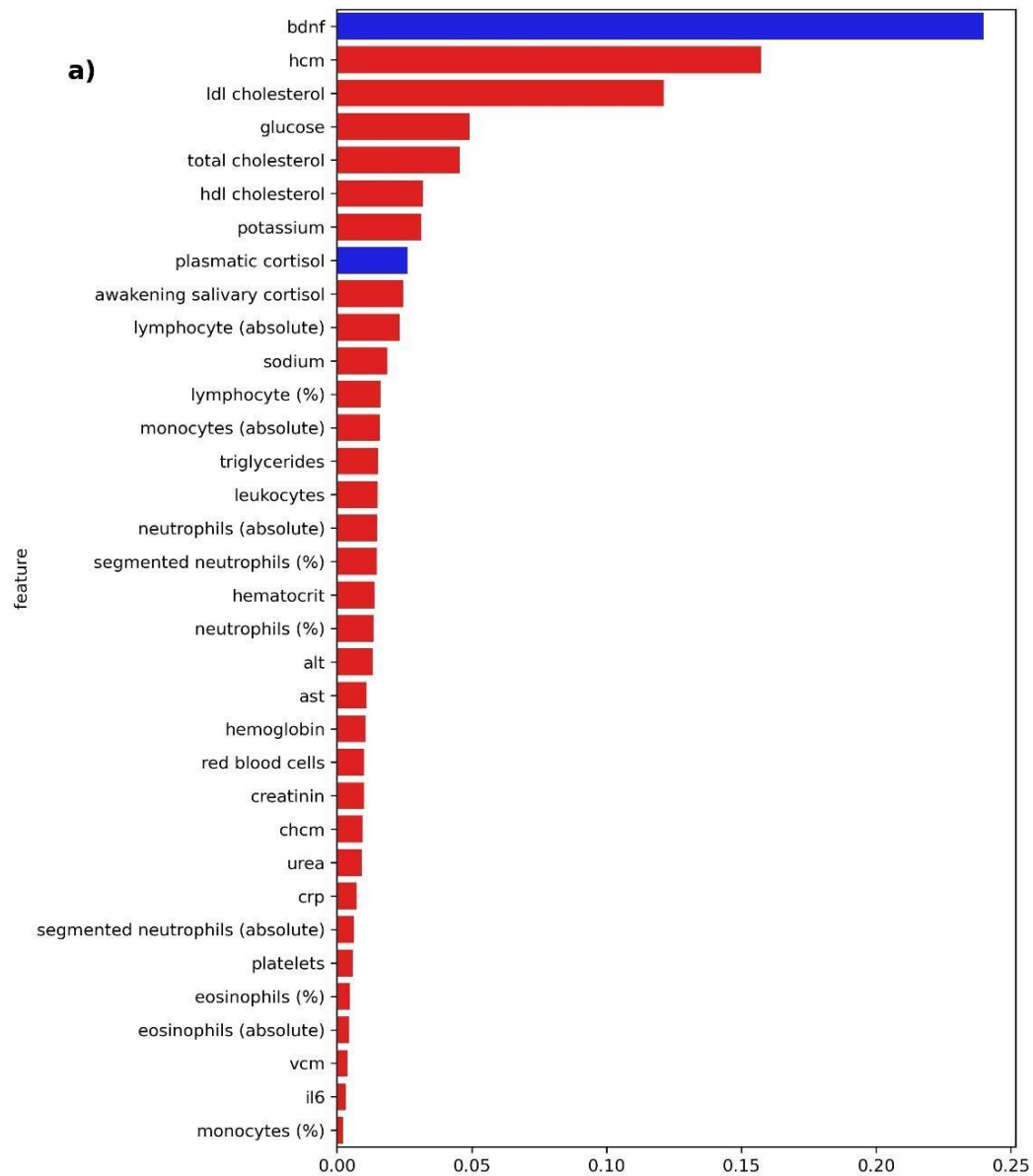


Figure 3. A) Feature importance ranking showing scores of biomarkers in random forest regression models predicting MADRS values. The blue bars represent features that when selected together improve models performance: BDNF and plasmatic cortisol. B) Pearson correlation of delta BDNF values with delta MADRS scale values, illustrating the relationship between these two variables in the context of treatment response. C) Pearson correlation of delta plasmatic cortisol values with delta MADRS scale values, illustrating the relationship between these two variables in the context of treatment response.

Discussion

In the present work, we developed a machine learning approach to identify significant blood and saliva biomarkers, defined here as features, with the aim of evaluating the significance of these biomarkers in depression and the influence an ayahuasca intervention has on them. As general main results, we observed that: (a) three main features stood out when differentiating healthy controls from patients with TRD: awakening salivary cortisol, C-reactive protein (CRP), and total cholesterol. In Wilcoxon comparisons, CRP and total cholesterol presented clear differences, while differences in salivary cortisol levels were marginally significant. Considering controls dosed with ayahuasca versus placebo (b), the models were inaccurate. In other words, the models did not find a difference between healthy individuals dosed with ayahuasca from those dosed with placebo, based on our selection of 31 distinct biomarkers. With respect to TRD patients dosed with ayahuasca versus placebo (c), the study identified several key biomarkers, including glucose, CRP, lymphocytes (both relative and absolute counts), AST, BDNF, red blood cells count, and LDL cholesterol, which had a significant impact on separating TRD patients dosed with ayahuasca or placebo. Finally, in (d) we evaluated the potential of features to predict the antidepressant effects measured by the MADRS scale, specifically for patients dosed with ayahuasca. Our results depicts an association between changes in two key biomarkers and improvements in depression symptoms, as reflected in MADRS scores. Specifically, when Brain-Derived Neurotrophic Factor (BDNF) levels increase, there tends to be an improvement in depression symptoms. However, when plasmatic cortisol levels increase, the improvement in depression symptoms is typically less pronounced or may even reverse, indicating a potential worsening of symptoms.

Several studies have corroborated our findings (Figure 1), reinforcing the relevance of certain biomarkers in distinguishing healthy subjects from patients with TRD. A study by Lamers et al. [23] revealed that the cortisol awakening response (CAR) in subjects diagnosed with melancholic depression, a subtype of major depressive disorder, was higher than those with atypical depression and controls, emphasizing a distinct CAR in control individuals versus those diagnosed with depression, which aligns with our findings of decreased awakening salivary cortisol in TRD patients compared to healthy. Our previous studies have shown that CAR level at baseline was lower for patients than healthy controls [15]. While several theories try to explain the etiology of the blunted awakening cortisol response, it seems that the best approach involves adrenal failure after a long-last period of overactivation [24,15]. It is possible that the low cortisol levels presented by TRD patients may be one of the causes of an inflammatory, depression-inducing profile since cortisol is a potent anti-inflammatory hormone [25].

Another study on the same cohorts [16] found that patients with TRD presented significantly higher levels of CRP than controls, which is consistent with our results. This study also found that increased severity of depressive symptoms was correlated with high levels of CRP, further supporting its role as a potential biomarker for TRD. Moreover, another study by Galvão et al. [17] found that patients with TRD had significantly lower levels of serum cortisol and CAR, and higher levels of mature brain-derived neurotrophic

factor (mBDNF) and CRP than controls. Besides, other studies observed that the total cholesterol levels circulating in peripheral blood are increased in depressive patients [27,28]. However, other works have shown an inverse correlation between total cholesterol and depression severity [26,27,28]. These observed changes could likely be attributed to the fact that our study group consisted of patients with TRD, a condition that may present unique biochemical profiles compared to other forms of depression. The elevated total cholesterol levels observed in patients with depression could be attributed to a combination of factors, including **a)** biochemical changes associated with depression that alter lipid metabolism [29], a common physiological response to stress-related disorders [30], **b)** behavioral aspects of depression such as sedentary lifestyle, and poor medication adherence [31], and **c)** the impact of certain antidepressants on lipid profiles [32]. While our findings are not entirely novel, they provide confirmation and extension of previous results, supporting the use of these three biomarkers as complementary tools in the diagnosis and treatment of TRD.

Secondly, we investigated the biochemical differences between healthy controls dosed with ayahuasca versus placebo. The metrics of our machine learning model indicated that ayahuasca had a similar impact on peripheral blood and saliva features in the control group dosed with ayahuasca compared to placebo. This finding is in line with previous studies that have reported the safety and tolerability of ayahuasca in controlled settings [33; 34]. These studies, along with our current findings, contribute to the growing body of evidence supporting the safety of ayahuasca intake in humans, further emphasizing the relevance of studies on its therapeutic benefit.

Next, we investigated the potential of biochemical features in separating patients with TRD dosed with ayahuasca from those patients dosed with placebo (Figure 2). Our results suggest that changes in eight key biomarkers, glucose, CRP, lymphocytes (relative count), lymphocytes (absolute count), AST, BDNF, red blood cells count, and LDL cholesterol, can successfully distinguish patients with TRD dosed with ayahuasca from those patients dosed with placebo. Additionally, in Wilcoxon tests, we observed a decrease in Red Blood Cell (RBC) count following ayahuasca treatment in both patients with TRD and healthy subjects (Supplementary Tables 2 and 3). In the TRD group, we noticed a slight decrease in RBC levels after treatment. This change was not present in the placebo group, indicating a potential physiological response to the treatment in the TRD group. In contrast, in the healthy subjects group, there was a clear decrease in RBC levels post-treatment. This change was not seen in the placebo-treated healthy individuals. The findings suggest that the influence of ayahuasca on different biological systems varies significantly between patients with TRD and healthy volunteers.

We found a potential influence of ayahuasca on RBC count, with a more pronounced effect observed in healthy subjects compared to TRD patients. Our findings align with those of Vizeli and Liechti [35], who observed a decrease in RBC count in women following MDMA administration. Although they attributed this decrease to the sampling of blood for pharmacokinetic analyses rather than the use of MDMA itself, it is plausible that both ayahuasca and MDMA could influence these changes in RBC count. Ayahuasca contains N,N-Dimethyltryptamine (DMT), a potent psychedelic substance that activates serotonin (5-HT)

receptors. Serotonin, a vasoactive amine, can cause vasoconstriction or vasodilation depending on the type of vessel and the 5HT receptor subtype activated. Changes in blood flow and pressure induced by DMT could indirectly affect RBC production or survival. Similarly, MDMA is known to increase the release of serotonin and other monoamines, which could lead to vasoconstriction and increased blood pressure, potentially affecting RBC production or survival. Finally, the ingestion of ayahuasca or MDMA is often associated with intense psychological experiences, which could trigger a stress response. This could potentially induce changes in various physiological parameters, including RBC count. It is important to note the difference in the time window of our study compared to the MDMA study[34]. In our study, the changes in RBC count were observed 48 hours after ayahuasca administration. In contrast, the MDMA study measured RBC count at the start and at the end of the study visits, which had variable time intervals (29 ± 22 days) after the last MDMA administration. This difference in timing could potentially influence the observed effects on RBC count and should be taken into account when comparing the results of both studies.

Additionally, our study builds upon the previous work of our research group, which reported a significant reduction in median CRP levels in patients with TRD following an ayahuasca intervention, suggesting a potential anti-inflammatory effect of ayahuasca. This aligns with a previous research that associated changes in CRP with ayahuasca's antidepressant effects[16]. Furthermore, our results resonate with observations that emotional and physiological parameters during an ayahuasca session were identified as critical moderators of major depression biomarkers, including CRP [36]. These collective findings underscore the potential role of ayahuasca in modulating inflammation, a key aspect in the pathophysiology of depression, reinforcing its therapeutic potential for depression.

Furthermore, our analysis identified glucose and lymphocytes as significant features for distinguishing treatment effects in TRD patients. Surprisingly, these biomarkers showed notable changes following dosing with the placebo, but not with ayahuasca. Median glucose levels significantly decreased after dosing with the placebo. Conversely, lymphocyte levels increased after dosing with the placebo, suggesting an immune response potentially triggered by the trial conditions. These unexpected findings highlight the complexity of physiological responses in clinical trials and underscore the potent influence of the placebo effect. The substantial changes observed in the placebo group reflect the power of expectation and belief in treatment efficacy, even when the treatment is inactive. This aligns with previous findings [12] where a strong placebo response was observed in the early stages of a trial investigating the antidepressant effects of ayahuasca. Furthermore, changes in glucose were not observed in the ayahuasca group, suggesting a possible modulation of ayahuasca on glucose.

Although AST, BDNF, and LDL did not individually show significant differences post-treatment, they were still selected as important features by our machine-learning model. This suggests that these biomarkers, in combination with the others selected, might be contributing to the overall ability of the model to distinguish between dosing with ayahuasca or placebo. The random forest analysis is well-suited to capture nonlinear relationships and complex interactions, allowing for a more comprehensive understanding of the multifaceted nature of the relationship between treatments and TRD patients. AST, an enzyme involved in liver

function, although not significantly altered, could be subtly influenced by ayahuasca, as changes [36] in liver function markers following ayahuasca intake have been reported. BDNF, a critical player in neuroplasticity and brain health, has been associated with major depressive disorder. While our study did not find significant changes in BDNF levels post-ayahuasca dosing, previous research has suggested a potential role of ayahuasca in modulating BDNF levels [14].

In addition, LDL cholesterol, a key factor in cardiovascular health, did not show significant changes in our study. However, the potential influence of ayahuasca on metabolic health [32], could indirectly affect LDL cholesterol levels. The random forest model, adept at capturing nonlinear relationships and complex interactions, allowed us to appreciate the multifaceted nature of the relationship between these treatments and the physiological responses in TRD patients. This highlights the potential of ayahuasca to modulate a complex network of biomarkers contributing to its therapeutic effects.

In the earlier parts of our study, BDNF and plasmatic cortisol were identified as the two most significant predictors of changes in MADRS scores following dosing with ayahuasca. BDNF demonstrated a strong negative correlation with delta MADRS scores, while plasmatic cortisol showed a strong positive correlation. This is in line with previously reported findings that ayahuasca administration increased BDNF levels and decreased cortisol levels in the blood of healthy volunteers[14]. However, when we compared the delta values for BDNF and plasmatic cortisol between the placebo and ayahuasca groups, we found that the differences were not statistically significant. This suggests that while BDNF and plasmatic cortisol may be associated with the antidepressant effects of Ayahuasca, their modulation by ayahuasca is not the sole determinant of these effects. This is consistent with reported findings that psychedelics induce a range of neuroplastic changes beyond the modulation of BDNF and cortisol[38].

The relationship between plasmatic cortisol and psychedelics, including isolated-DMT and ayahuasca, is complex and multifaceted. For instance, it has been found that ayahuasca decreased cortisol levels, suggesting it might have a modulating effect on the hypothalamic-pituitary-adrenal (HPA) axis, which is responsible for the body's stress response and cortisol production[39]. Another study found that ayahuasca induced acute increases in cortisol levels, but these increases were not associated with the intensity of the subjective effects of ayahuasca, suggesting that the relationship between ayahuasca and cortisol might be influenced by other factors[15]. In the context of our study, the significant predictive role of plasmatic cortisol in the antidepressant effects of ayahuasca could be related to its potential modulating effects on the HPA axis. Dysregulation of the HPA axis and elevated cortisol levels have been implicated in the pathophysiology of depression, and the ability of ayahuasca to modulate cortisol levels might contribute to its therapeutic effects.

Recent studies have further expanded our understanding of the neuroplastic effects of psychedelics. For instance, it has been reported that psychedelics promote neuroplasticity through the activation of intracellular 5-HT_{2A} receptors[40,41]. Similarly, it was found that a single dose of 5-MeO-DMT stimulates cell proliferation, neuronal survivability, and morphological and functional changes in the adult mice ventral dentate gyrus[43]. There have

also been reported short-term changes in the proteome of human cerebral organoids induced by 5-MeO-DMT[13]. These findings underscore the potential of psychedelics in inducing neuroplastic changes and their therapeutic effects.

Our results, together with these findings, suggest that the antidepressant effects of ayahuasca and other psychedelics may be mediated by a complex interplay of neuroplastic changes. These changes could involve not only the modulation of BDNF and plasmatic cortisol but also other mechanisms such as the modulation of mammalian target of rapamycin (mTOR) and early growth response-related pathways [40, 41], as well as alterations in cortical network activity [42].

Limitations and perspectives

While our study provides insights into the potential biomarkers of the effects of ayahuasca in different biological systems, both in treatment in TRD patients as well as in healthy individuals, it is important to acknowledge a few limitations and caveats. The small sample size may limit the generalizability of our findings, and larger studies are needed to validate our results. Our machine learning approach, while powerful and innovative in the context of clinical trials, is dependent on the quality and quantity of the input data. Specifically, the Recursive Feature Elimination (RFE) method, while effective in identifying key biomarkers, may overlook important interactions between less influential features. Therefore, combining machine learning techniques with traditional statistical analyses could provide a more comprehensive understanding of biomarker relationships in mental health research. The controlled clinical setting of our study may not fully reflect real-world conditions, and the effects of ayahuasca may vary in different contexts. Additionally, our study's short time window of 48 hours limits the range of observable changes in different biomarkers post-dosing. Future studies could benefit from a longer follow-up period to capture potential delayed effects of ayahuasca. Lastly, while our study supports the safety of ayahuasca intake in healthy individuals, further research is needed to investigate its long-term effects, especially in individuals with pre-existing mental health conditions.

Conclusions

This study utilized machine learning to analyze data from a randomized controlled trial on treatment-resistant depression (TRD) and the effects of ayahuasca in patients with TRD and healthy individuals, giving support to the importance of investigating different biomarkers that might be related to the effects of psychedelics and their safe therapeutic use.

Ethics declarations

The study was approved by the HUOL Research Ethics Committee (# 579.479). All subjects provided written informed consent before participation. This study was registered at ClinicalTrials.gov (NCT02914769).

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Declaration of conflicts

The author(s) have no conflicts of interest to declare.

Data availability

The datasets used and/or analyzed during the current study are available upon reasonable request from the corresponding author. The scripts for Python and R used in this study can be accessed at: github.com/isaacbcampos/mlpaperaya.

Declaration of conflicts

The author(s) have no conflicts of interest to declare.

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Contributions

I.C.B., E.A., S.M.R., F.P.F. and D.B.A. conceived the project and the overall study design. I.G.M. and I.C.B. designed the machine learning approach and its application to the acquired data. I.C.B. and I.G.M. developed the Python scripts for the machine learning model. I.C.B. and F.P.F. handled the acquisition of data, and their formal analysis. B.L.S., N.G.C., F.P.F. and D.B.A. validated the results and analysis. I.C.B., E.A., S.M.R., and D.B.A. D.B.A. and F.P.F. supervised, administered, and acquired funding for the project as part of the Brain Institute at the Federal University of Rio Grande do Norte. I.C.B. drafted the original manuscript and all authors were involved in reviewing and revising the final manuscript.

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